Peter Avgerinos

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An ambitious Engineer with experience in backend development and deployment of large scale AI systems and applications, and in building compiler tooling such as parsers, LSPs, and static analysis passes in MLIR. Currently pursuing a MSc at Imperial College London in Computational Science and HPC.

EXPERIENCE

Compilers Lab - UFMG *Belo Horizonte, Brazil*

Master Thesis Project *May 2026 - Current*

- Built compiler infrastructure, under the supervision of Professor Fernando Pereira, that reformulates the multidirectional **Sparsity Propagation Analysis (SPA) as a static analysis within the MLIR framework**, targeting the *linalg* dialect.
- Achieved **100% forward-pass operator coverage** across a real-world evaluation set of **15 ONNX vision models** (ResNet, VGG, etc.). Implemented and validated transfer functions for all 16 distinct MLIR ops (matmul variants, conv2d variants etc.) that are apparent in the dataset.
- Built a **model-ingestion pipeline, using Docker** for reproducibility, that turns ONNX vision models into MLIR test inputs, and measures the amount of sparsity inferred by the analysis, as well as how far proven sparsity propagates.
- Achieved **over 88% test coverage** with a test suite of **59 FileCheck-based lit tests + 22 GoogleTest unit tests** covering every pass and op visitor, and orchestrated **GitHub Actions CI for building, testing, coverage and formatting**.

Vertical Playground *Athens, Greece*

Independent Researcher *Feb 2025 - Current*

- Led the development of **Erminia, a Domain Specific Language,** that allows Humans and LLMs to interpret ARC-AGI Abstracted Images as code, built in **Rust**. The Erminia DSL is a collection of software tools and libraries (parser, AST, LSP, Diagnostics) designed to facilitate research in the ARC-AGI space.
- Building a **static analysis pass** for **propagating error bounds** when using **Stochastic Rounding in long sum operations** within the **MLIR framework**. This pass allows for assessing the unit roundoff error accumulation inside the IR through error propagation analysis, enabling compile-time assessment of Stochastic Rounding hardware benefits, reducing the need for trials of hardware or software simulation.

Institute of Communication and Computer Systems (ICCS) *Athens, Greece*

Software Engineer *Oct 2024 - Aug 2025*

- Designed and developed an **LLM-powered recommendation system** for an e-commerce client, functioning as an agent capable of product search, comparison, and checkout based on classified user intent. Stack included OpenAI API, ElevenLabs API, LangChain, fine-tuned Transformers trained on synthetic data, pgvector on Azure Postgres for semantic search, and Logfire for observability. This system enabled automated user-journeys within the client application.
- Led technical writing and consortium coordination for Horizon Europe and European Defense Fund proposals across healthcare, smart cities, defense, and cybersecurity, participating in 5 proposal submissions in total.

Intern *July 2024 - Sep 2024*

- Designed and developed an XAI-driven cognitive assessment tool, named DEMET (DEMentia Explainable Transformer) with PyTorch and Scikit-learn, which leveraged ensemble learning techniques in order to combine statistical classifiers and transformers to detect dementia from spontaneous speech samples. DEMET demonstrated that ensemble models can significantly improve performance, **achieving over 97% accuracy on the DementiaBank dataset**.

EDUCATION

MSc Candidate in Computational Science and Engineering *2025-2026*
Imperial College London, London UK

BEng, MEng in Electrical and Computer Engineering *2019-2024*
National Technical University of Athens, Athens Greece

Certification in CUDA and GPU Programming *2026*
Erlangen National High Performance Computing Center, Erlangen Germany

TECHNICAL SKILLS

Languages	C++, C, Python, Rust
Backend Development	Postman, LangChain, PyTorch, Hadoop, Spark, FastAPI, MySQL, PostgreSQL, Pydantic, Logfire
Testing & Benchmarking	Pytest, GoogleTest, GoogleBench, Lit, Hyperfine
Accelerated Computing	OpenMP, MPI, CUDA
BuildSys/Cloud/DevOps	Makefile, CMake, Cargo, uv, Docker, Git, Github Actions, Vim, Azure, AWS, NGINX
Compilers/Parsers/OS	Bison, Flex, LLVM, MLIR, QEMU/KVM, Linux Kernel API & VFS